

Reinforcement learning for shaken lattice

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1 Introduction

We have applied deep reinforcement learning to the control problem for the design of the shaken lattice experiment using the elements illustrated in Figure 1(a).

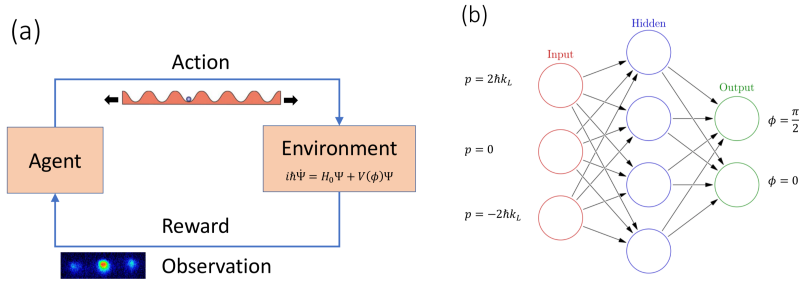


Figure 1: (a) The elements of reinforcement learning. (b) The topology of the deep-Q network.

In this setting, the observed state is the momentum population distribution, the actions are the phases of the optical lattice that determine the positions of the nodes and antinodes, and the reward is based on the fidelity between the quantum state and a specified target quantum state. The decision process is segmented into discrete and equal time steps, and the environment evolves according to the Schrödinger equation. We have implemented a single-layer deep-Q network as the “brain” of the agent for selecting actions. The deep-Q network takes the momentum population distribution as input, and subsequently outputs Q-values for the discrete set of phase choices. The Q-values evaluate the expected rewards for possible actions conditioned on the input state, and most often we simply select the action with maximum Q-value. Since global exploration is also essential in reinforcement learning as a way to avoid local minima, with small probability ϵ we select random actions. At each time step, a sample that consists of the input state, the action chosen, the next state, and the reward, (s, a, s', r) , is generated and stored in a replay buffer. We sample a batch from the replay buffer in order to calculate the loss, and this allows us to update

the parameters in the deep-Q network in the direction of decreasing loss, i.e. $\lambda \leftarrow \lambda - \alpha \nabla_{\lambda} L$, where λ represents the set of network parameters, and α is the learning rate.

When the network is properly trained, it should follow the Bellman equation,

$$Q(s, a) = r + \gamma \max_{a'} Q(s', a'), \quad (1)$$

where γ is the discount rate, and represents how much the future rewards matter. In order to establish the Bellman equation, the most common choice of loss function is

$$L = \frac{1}{|B|} \sum_i^{|B|} \left[Q(s_i, a_i) - (r_i + \gamma \max_{a'} Q(s'_i, a')) \right]^2, \quad (2)$$

where $|B|$ is the batch size.

An example implementation of this consists of a network with populations in 7 momentum states,

$$\{|2n\hbar j_L\rangle \mid n = -3, -2, -1, 0, 1, 2, 3\},$$

as input, 32 hidden nodes, and 5 Q-values for the set of discrete phase values

$$\{n\pi/4 \mid n = -2, -1, 0, 1, 2\},$$

as output. The output y and the input x of the network are related as follows,

$$h = \text{ReLU}(W_1 x + b_1) \quad (3)$$

$$y = \text{ReLU}(W_2 h + b_2), \quad (4)$$

where W_1 , W_2 are the weights and b_1 , b_2 are the biases. The topology of the network is shown in Figure 1(b). To obtain a control strategy for the splitting stage of the interferometer, we start from the ground state of the system, and specify as the target quantum state the first-excited Bloch state. Similarly, for recombination, we start from the first-excited Bloch state, and specify the ground state as the target. The reward is chosen to be

$$r = 2 * (\Delta F > 0) + (F > 0.4) * 0.2 * F + (F > 0.7) * 2 * F + (F > 0.9) * 20 * F, \quad (5)$$

where F is the fidelity and ΔF is the difference between the fidelity at the previous step and the current fidelity. Figure 2 shows the results for both cases after training for a hundred episodes. The agent has discovered strategies that achieve $\sim 95\%$ fidelity.

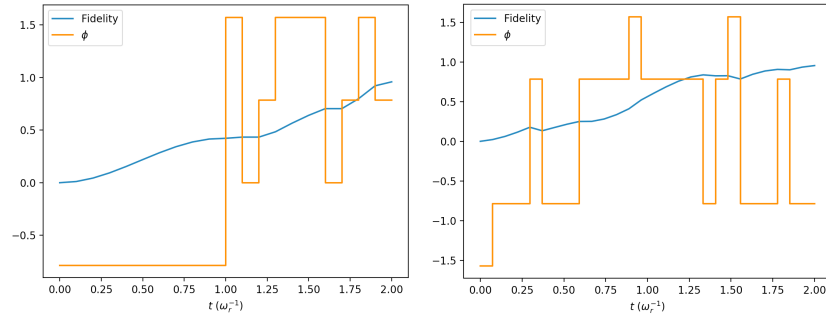


Figure 2: Fidelity and control phase as functions of time. Left: Result for splitting. The fidelity is compared to the $N = 1$ Bloch state. Right: Result for recombination. The fidelity is compared to the $N = 0$ Bloch state.